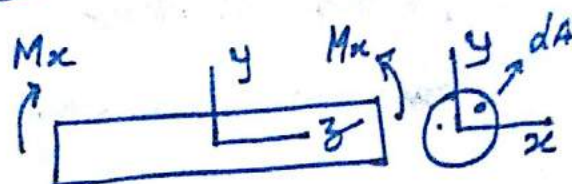


Shear flow in Open Sections - UNIT - 2

Unsymmetrical Bending:



From the bending equation

$$-\frac{\sigma_z}{y} = \frac{M_x}{I_{xx}} = \frac{E}{R} \quad \text{--- (1)}$$

Now, let us consider the Moment M_y , which is

$$M_y = \int_A \sigma_z \cdot x \, dA$$

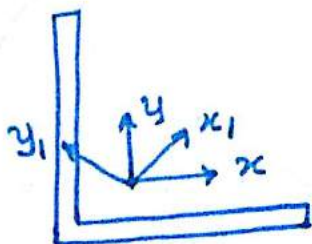
$$M_y = -\frac{M}{I_{xx}} \int_A y \cdot x \cdot dA$$

Now, we can recognise the term $\int_A y \cdot x \cdot dA$ which is the product moment of inertia.

$$M_y = -\frac{M}{I_{xx}} \cdot I_{xy}$$

and for a Symmetrical section $I_{xy} = 0$. and for Unsymmetrical section $I_{xy} \neq 0$.

for an Unsymmetrical section, we need to obtain a plane which will help us make equation (1) useable by making $I_{xy} = 0$.



y_1 and x_1 are called principal directions

Hence, we can say for an Unsymmetrical bending the direct stress

$$\sigma_x = \frac{M_x (I_{yy} \cdot y - I_{xy} \cdot x)}{I_{xx} I_{yy} - I_{xy}^2} + \frac{M_y [I_{xx} \cdot x - I_{xy} \cdot y]}{I_{xx} I_{yy} - I_{xy}^2}$$

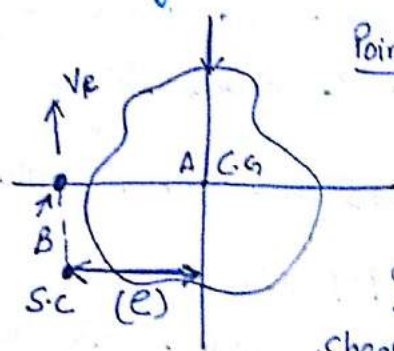
Shear Centre:

(2)

In aerospace structures we generally have thin members which are asymmetrical, where the thickness of the member is small when compared to the longitudinal dimensions. Generally, the thin walled structures are very good in bending but weak in torsion.

- When the unsymmetric cross section (i.e., the axis of transverse load or line of action does not coincide with the C.G., Centroid or the mass center of the cross section, additional moments are induced, which create torsional effects during bending.

Hence, shear centre is defined as, "The point about which when loaded, does not produce any torsional effect in bending". i.e. pure bending happens, when loaded about that point and when loaded torsionally about that point, there is no bending induced.



Point - A :-

C.G. is the point where the self weight loads act

Point B :-

Shear Centre is the point where the transverse loads are acting.

Shear Centre may lie within the body or outside the body.

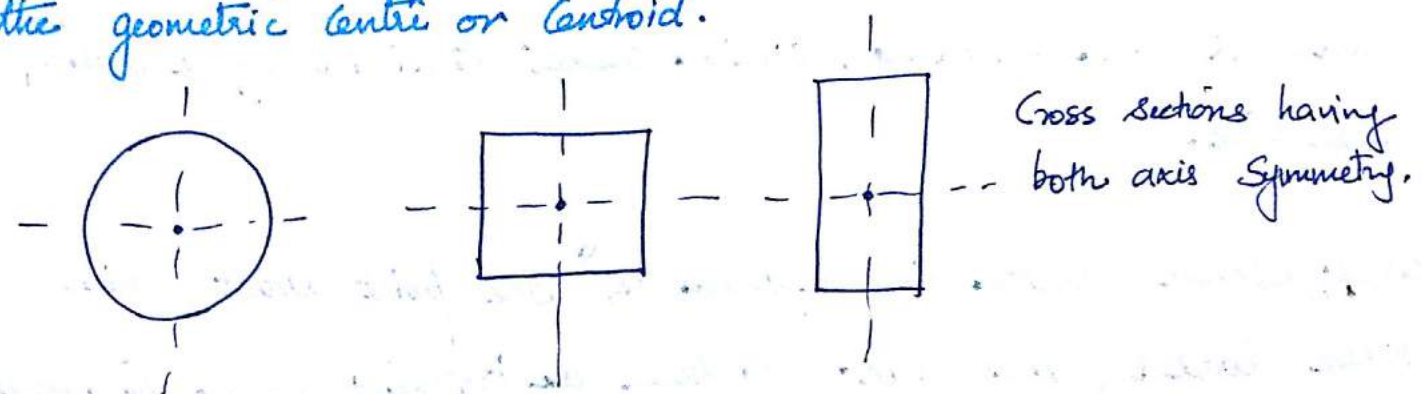
Now, the reaction of all forces act at shear centre, (V_y).

The moment which is due to the reaction of forces is $V_y \cdot e = M$.

Now, for static equilibrium we know, that V_f has to be equal to the applied shear. Hence, to calculate the moment, we need the value of 'e'. For sections what should be the value of 'e'?

'e' - Offset of the Shear Centre from the Centroid.

Case I: When the Cross section has both axes of Symmetry, then the Shear Centre will coincide with the geometric centre or Centroid.



Case II. When there is one axis of Symmetry, the Shear Centre will lie on that axis of Symmetry, but it could lie anywhere on that axis of Symmetry.

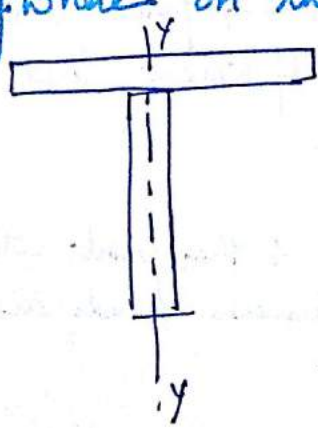


Fig. Shows a T-Section with one axis Symmetry about 'y-y' axis.

The Shear Centre will lie on that axis 'y-y' but could lie anywhere on that 'y-y' axis.

In this section, we will deal only with open sections bearing bending loads and not torsional loads.

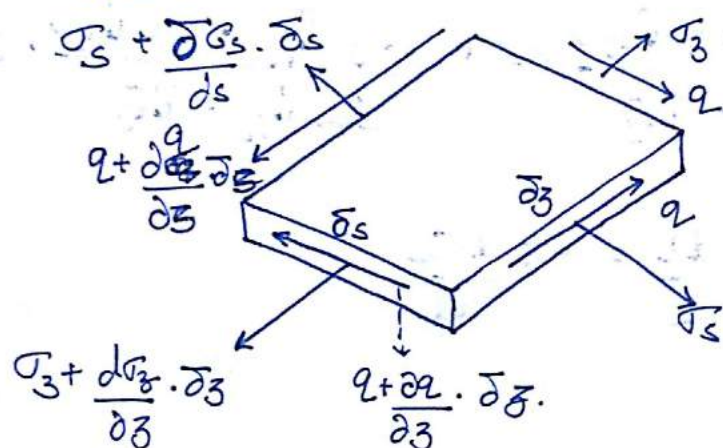
Hence, we can say that shear flow in a thin walled beam's.

$$q = \tau t$$

Where q is the shear flow and

τ is the shear stress &

t is the thickness which is constant over the length.



DIRECT stress &
SHEAR flow system on
the element.

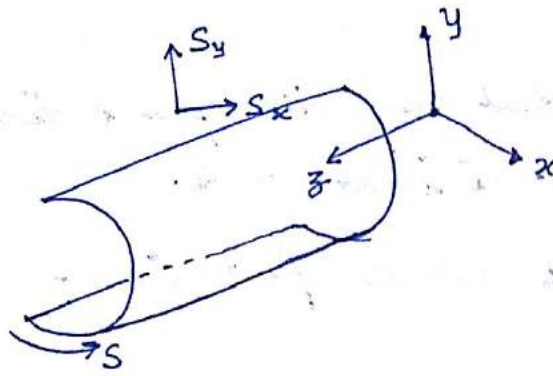
For equilibrium in the 's' direction.

$$\frac{\partial q}{\partial z} + t \frac{\partial \tau_s}{\partial s} = 0 \quad \text{or} \quad \frac{dq}{dz} + t \frac{d\tau_z}{dz} = 0$$

We know from unsymmetrical bending,

$$\tau_z = \frac{(M_y I_{xx} - M_x I_{xy})x}{I_{xx} I_{yy} - I_{xy}^2} + \frac{(M_x I_{yy} - M_y I_{xy})y}{I_{xx} I_{yy} - I_{xy}^2}$$

$$\frac{d\tau_z}{dz} = \frac{[(\partial M_y / \partial z) I_{xx} - (\partial M_x / \partial z) I_{xy}]}{I_{xx} I_{yy} - I_{xy}^2} + \frac{[(\partial M_x / \partial z) I_{yy} - (\partial M_y / \partial z) I_{xy}]}{I_{xx} I_{yy} - I_{xy}^2} \cdot y$$



From relationship of shear force and bending moments,

$$\frac{\partial M_y}{\partial z} = S_x$$

Hence,

$$\frac{d\sigma_z}{dz} = \frac{(S_x I_{xx} - S_y I_{xy})}{I_{xx} I_{yy} - I_{xy}^2} x + \frac{(S_y I_{yy} - S_x I_{xy})}{I_{xx} I_{yy} - I_{xy}^2} y$$

$$\frac{dq}{ds} + t \frac{d\sigma_z}{dz} = 0, \quad \frac{dq}{ds} = -t \frac{d\sigma_z}{dz}$$

Substituting, we get

$$\frac{dq}{ds} = - \frac{(S_x I_{xx} - S_y I_{xy})}{I_{xx} I_{yy} - I_{xy}^2} t x - \frac{(S_y I_{yy} - S_x I_{xy})}{I_{xx} I_{yy} - I_{xy}^2} t y$$

$$\int_0^s \frac{dq}{ds} ds = - \frac{(S_x I_{xx} - S_y I_{xy})}{I_{xx} I_{yy} - I_{xy}^2} \int_0^s t x ds - \frac{(S_y I_{yy} - S_x I_{xy})}{I_{xx} I_{yy} - I_{xy}^2} \int_0^s t y ds$$

$$q_s = - \frac{(S_x I_{xx} - S_y I_{xy})}{I_{xx} I_{yy} - I_{xy}^2} \int_0^s t x ds - \frac{(S_y I_{yy} - S_x I_{xy})}{I_{xx} I_{yy} - I_{xy}^2} \int_0^s t y ds$$

When we have one axis of symmetry.

$$q_s = - \frac{S_x}{I_{yy}} \int_0^s t x ds - \frac{S_y}{I_{xx}} \int_0^s t y ds$$

When there is axis of symmetry about x-axis then the equation reduces to

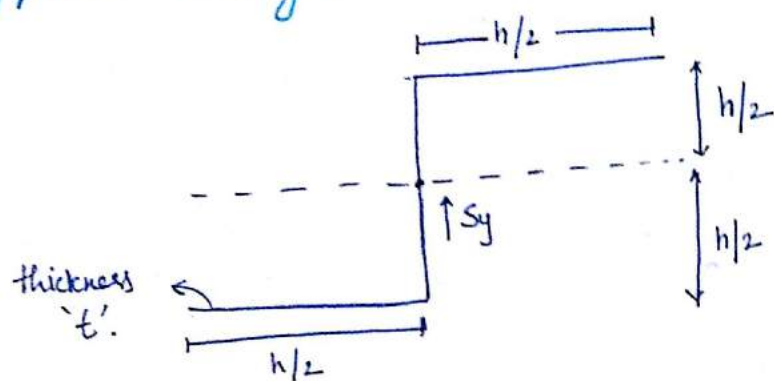
$$Q_s = -\frac{S_y}{I_{xx}} \int_0^s t \cdot y \cdot ds$$

because, $S_x = 0$, and $I_{xy} = 0$. and similarly we can write the equation as the following for axis of symmetry for y -axis, as

$$Q_s = -\frac{S_x}{I_{yy}} \int_0^s t \cdot x \cdot ds.$$

Problems:

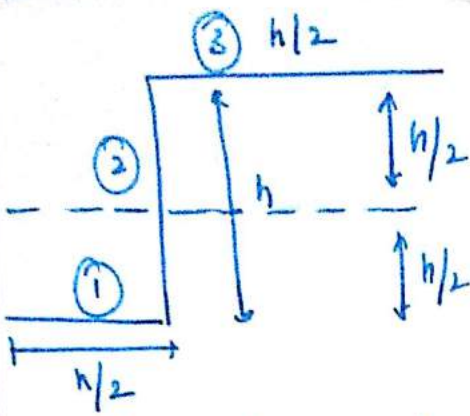
Determine the shear flow distribution in the thin walled Z-section as shown in the figure. Due to the shear load applied through the shear centre of the section.



Sol:

In the given problem $S_x = 0$.

We know that due to symmetry about x and y axis the shear centre will lie within the centroid of the cross section.



Moments of Inertia is given by.

$$I_{xx} = \sum (I_{xx} + Ay^2)$$

$$I_{yy} = I_{yy} + Ay^2$$

Calculating the moment of Inertia,

$$I_{xx}]_1 = \frac{h}{2} \times \frac{t^3}{12} = \frac{ht^3}{24} \quad [\text{For 1st segment}]$$

$$I_{xx}]_2 = t \times \frac{h^3}{12} = \frac{h^3t}{12} \quad [\text{For 2nd segment}]$$

$$I_{xx}]_3 = \frac{h}{2} \times \frac{t^3}{12} = \frac{ht^3}{24} \quad [\text{For 3rd segment}]$$

$$\text{Area for 1st segment } A_1 = \frac{h}{2} \times t = ht/2$$

$$A_2 = ht$$

$$A_3 = ht/2$$

$$I_{xx} = \left[\frac{ht^3}{24} + \left(\frac{ht}{2} \right) \left(\frac{h}{2} \right)^2 \right]_1 + \left[\frac{h^3t}{12} + ht \cdot (0) \right]_2 + \left[\frac{ht^3}{24} + \left(\frac{ht}{2} \right) \left(\frac{h}{2} \right)^2 \right]_3$$

Neglecting higher order terms, Cause, $t \ll h$.

$$\therefore I_{xx} = \frac{h^3t}{8} + \frac{h^3t}{12} + \frac{h^3t}{8} =$$

Simplifying

$$I_{xx} = h^3t/3 \quad //$$

$$I_{xx} = h^3t/3$$

$$I_{yy_1} = t \times \left(\frac{h^3}{2 \times 12} \right) = \frac{th^3}{96} \text{ which is also equal to body 3.}$$

(8)

$$I_{yy_1} = I_{yy_3} = \frac{th^3}{96}$$

$$I_{yy_2} = \frac{ht^3}{12}$$

$$\text{Hence, } I_{yy} = \left[\frac{th^3}{96} + \frac{ht}{2} \left(\frac{h}{4} \right)^2 \right]_1 + \left[\frac{ht^3}{12} + ht(0) \right]_2 + \left[\frac{th^3}{96} + \frac{ht}{2} \left(\frac{h}{4} \right)^2 \right]$$

Neglecting higher Order terms, we get.

$$I_{yy} = \frac{th^3}{96} + \frac{th^3}{32} + \frac{th^3}{96} + \frac{th^3}{32} = \boxed{\frac{h^3 t}{12} = I_{yy}}$$

Computing the cross product of Inertia,

$$I_{xy}]_1 = A_1(x-\bar{x})(y-\bar{y}) = \frac{ht}{2} \left(\frac{h}{4} - \frac{h}{2} \right) \left(\frac{h}{2} - \frac{h}{2} \right)$$

$$I_{xy}]_1 = \frac{h^3 t}{16}$$

$$I_{xy}]_2 = A_2(x-\bar{x})(y-\bar{y}) = 0 \text{ as it coincides,}$$

$$I_{xy}]_3 = A_3(x-\bar{x})(y-\bar{y}) = \frac{ht}{2} \left(\frac{h}{2} - \frac{h}{4} \right) \left(\frac{h}{2} \right) = \frac{h^3 t}{16}$$

$$I_{xy} = I_{xy}]_1 + I_{xy}]_2 + I_{xy}]_3 = \frac{h^3 t}{16} + 0 + \frac{h^3 t}{16} = \frac{h^3 t}{8}$$

$$\text{Hence, } \boxed{\begin{aligned} I_{xy} &= \frac{h^3 t}{8} \\ I_{xx} &= \frac{h^3 t}{3} \\ I_{yy} &= \frac{h^3 t}{12} \end{aligned}}$$

Hence to find the shear flow due to the applied shear S_y through the shear centre. (9)

$$\text{Shear flow } q_s = \frac{S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \int_0^s t \cdot x \cdot ds - \frac{S_y I_{yy}}{I_{xx} I_{yy} - I_{xy}^2} \int_0^s t \cdot y \cdot ds.$$

We have neglected the S_x terms as $S_x = 0$,

$$q_s = \frac{S_y \left[\frac{h^3 t}{8} \right]}{\frac{h^3 t}{3} \cdot \frac{h^3 t}{12} - \left(\frac{h^3 t}{8} \right)^2} \int_0^s t \cdot x \cdot ds - \frac{S_y \left[\frac{h^3 t}{12} \right]}{\frac{h^3 t}{3} \cdot \frac{h^3 t}{12} - \left(\frac{h^3 t}{8} \right)^2} \int_0^s t \cdot y \cdot ds.$$

(Negative sign gets cancelled).

further simplification. ('t' gets cancelled out).

$$\rightarrow q_s = \frac{S_y [h^3/8]}{\frac{h^6}{36} - \frac{h^6}{64}} \int_0^s x \cdot ds - \frac{S_y [h^3/12]}{\frac{h^6}{36} - \frac{h^6}{64}} \int_0^s y \cdot ds$$

$$\rightarrow q_s = \frac{S_y}{h^3} \int_0^s 10.28x \cdot ds - \frac{S_y}{h^3} \int_0^s [6.85y] ds$$

$$q_s = \frac{S_y}{h^3} \int_0^s [10.28x - 6.85y] ds$$

Considering the bottom flange: $y = -h/2$, $x = -h/2 + s$

$$q_s = \frac{S_y}{h^3} \int_0^s [10.28(-h/2 + s) - 6.85(-h/2)] ds$$

and s varies from 0 to $h/2$.

$$q_s = \frac{S_y}{h^3} \int_0^{h/2} \left[-\frac{10.28h}{2} + 10.28s + \frac{6.85h}{2} \right] ds$$

$$q_s = \frac{S_y}{h^3} \int_0^{h/2} [10.28s - 1.715h] ds$$

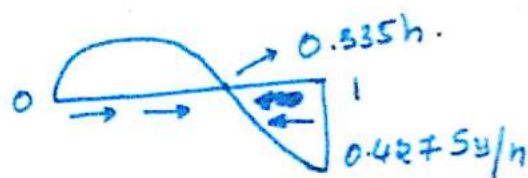
(10)

$$q_s = \frac{S_y}{h^3} \left[10.28 \frac{s^2}{2} - 1.715 S h \right]_{0}^{S=h/2} = \frac{S_y}{h^3} \left[5.14 s^2 - 1.715 S h \right]$$

at body ①, 

at ① $S=0$

$$q_1 = \frac{S_y}{h^3} [0]$$



at ② $S = h/2$.

$$q_2 = \frac{S_y}{h^3} \left[5.14 \frac{h^2}{4} - 1.715 \frac{h^2}{2} \right] = \frac{S_y}{h^3} [0.4275 h^2] = \frac{S_y}{h} [0.4275]$$

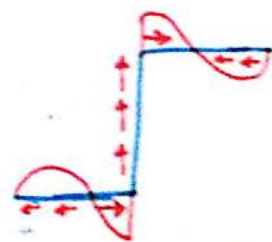
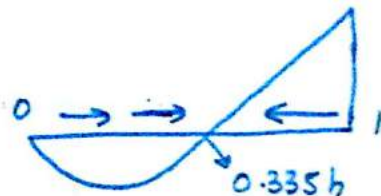
$$q_s = \frac{S_y}{h^3} [5.14 s^2 - 1.715 S h] \rightarrow \text{parabolic equation.}$$

$$\text{hence } 5.14 s^2 - 1.715 S h = 0$$

$S = 0.335 h$ indicates the sign-changes at

$0.335 h$.

Similarly, the top flange,



At the web, $x=0$, $y = -h/2 + s$, Varying from $0-h/2$ at 2 to 3

$$q_s = \frac{S_y}{h^3} \left[\int_0^s [10.28 x - 6.85 y] ds + q_2 \right] \quad (\text{because of load from flange})$$

$$q_{23} = \frac{S_y}{h^3} \int_0^s [10.28(0) - 6.85(-h/2 + s)] ds + q_2$$

$$q_{23} = \frac{S_y}{h^3} \int_0^s [3.425 h - 6.85 s] ds + q_2$$

$$q_{23} = \frac{S_y}{h^3} [3.425 h s - 3.425 s^2]_0^s + q_2$$

~~q_{23}~~

At $s=0$, for $q_{28} = \frac{S_y}{h^3} [3.425hs - 3.425s^2 + 0.425h^2]$ (11)

Applying limits

At $s=0$, at 2

$$q_3 = \frac{S_y}{h^3} [0.425h^2] = \frac{S_y}{h} [0.425]$$

at $s=h$,
at 3 $q_{23} = \frac{S_y}{h^3} [0.425h^2] = \frac{S_y}{h} [0.425]$

At the mid-section of the web,

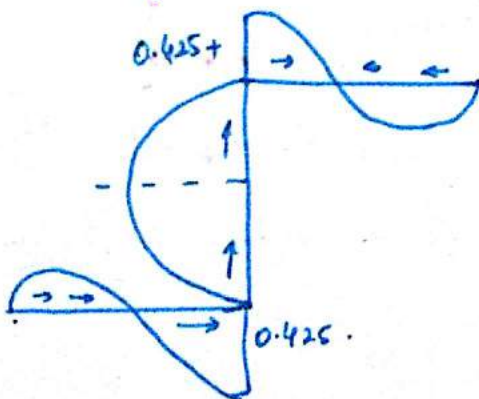
when $B = h/2$

$$q_{23} = \frac{S_y}{h^3} [\cancel{3.425hs} \quad 3.425 \frac{h^2}{2} - 3.425 \frac{h^2}{4} + 0.425h^2]$$

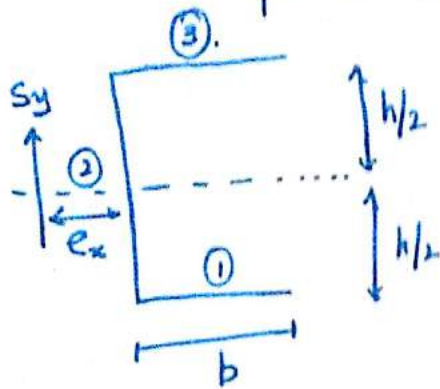
$$q_{23} = \frac{S_y}{h^3} [0.856h^2 + 0.425h^2]$$

$$q_{23} = \frac{S_y}{h^3} [1.28h^2]$$

$$q_{23} = \frac{S_y}{h} [1.28] \text{ at mid span.}$$



Problem: Calculate the position of the shear centre for a thin walled channel as shown in the fig. (12)



Since the beam is symmetric about x-axis and $S_x = 0$, we can say $I_{xy} = 0$.

Hence, the equation

$S_x = 0$ and $I_{xy} = 0$.

$$q_s = \frac{-(S_x I_{xx} - S_y I_{xy})}{I_{xx} I_{yy} - I_{xy}^2} \int_0^s t \cdot x \cdot ds$$

$$- \frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \int_0^s t \cdot y \cdot ds.$$

Hence, we have,

$$q_s = - \frac{S_y I_{yy}}{I_{xx} I_{yy}} = - \frac{S_y}{I_{xx}} \int_0^s t \cdot y \cdot ds \quad \text{--- (1)}$$

$$I_{xx1} = \frac{bt^3}{12} = I_{xx3}, \quad I_{xx2} = \frac{th^3}{12}$$

$$A_1 = bt = A_3, \quad A_2 = h \cdot t$$

$$I_{xx} = \sum I_{xx} + A y^2 \quad \left| \begin{array}{l} \text{Centroid calculation.} \\ \frac{\sum A \cdot x}{\sum A} = \bar{x}, \quad \frac{\sum A \cdot y}{\sum A} = \bar{y}, \end{array} \right.$$

$$I_{xx} = \left[\frac{bt^3}{12} + tb \left(\frac{h}{2} \right)^2 \right]_1 + \left[\frac{bt^3}{12} + tb \left(\frac{h}{2} \right)^2 \right]_3$$

$$+ \left[\frac{th^3}{12} + ht(0) \right]_2$$

$$I_{xx} = \frac{tbh^2}{4} + \frac{tbh^2}{4} + \frac{th^3}{12} = \frac{tbh^2}{2} + \frac{th^3}{12} = th^2 \left[\frac{b}{2} + \frac{h}{12} \right]$$

$$I_{xx} = \frac{th^2}{2} \left[b + \frac{h}{6} \right], \quad I_{xx} = \frac{th^2}{2} \left[\frac{6b+h}{6} \right] = \frac{th^3}{12} \left[\frac{6b}{h} + 1 \right] //$$

Shear flow $q_s = -\frac{S_y}{I_{xx}} \int_0^s t(y) \cdot ds$

(13)

q_s is in the flange.

$$q_s = -\frac{S_y}{I_{xx}} \int_0^s t(-h/2) ds = \frac{S_y}{2 I_{xx}} \int_0^s t h \cdot ds$$

$$q_s = \left[\frac{S_y}{2 I_{xx}} t h s \right]_0^s = \frac{S_y}{2 I_{xx}} t h \cdot s$$

$$q_{12} = \frac{S_y \cdot t \cdot h \cdot s}{2 I_{xx}}$$

Considering the moments along the axis of symmetry and $\Sigma M = 0$ about centroid lying on the axis of symmetry.

$S_y \cdot e_x = 2 \times \text{Moment from flange.}$

$$S_y \cdot e_x = 2 \int_0^b q_{12} (h/2) ds$$

$$S_y \cdot e_x = 2 \int_0^b \frac{S_y \cdot t \cdot h \cdot s}{2 I_{xx}} \cdot \left(\frac{h}{2}\right) \cdot ds \quad S_y \cdot e_x = \frac{S_y t \cdot h^2}{2 I_{xx}} \left[\frac{s^2}{2} \cdot ds \right]_0^b$$

$$e_x = \frac{t h^2}{2 \left[\frac{t h^3}{3} \cdot \left(\frac{6b}{h} + 1 \right) \right]} \cdot \frac{b^2}{2} = \frac{3 t h^2 b^2}{t h^3 \left[\frac{6b}{h} + 1 \right]} = \frac{3 b^2}{\left[\frac{6b}{h} + 1 \right] \cdot h}$$

Hence, we can say.

$$e_x = \frac{3 b^2}{h \cdot \left[\frac{6b}{h} + 1 \right]}$$

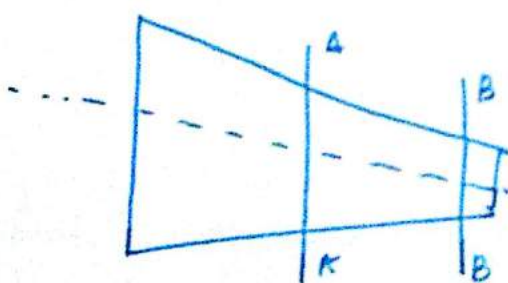
Shear Centre (Theory).

14

- ① Shear Centre - a point where shear load produces no twisting.
- ② By reciprocal theorem, ~~for~~ it is also a point about which ~~can~~ when twisted produces pure torsion and no bending.
- ③ Shear Centre for a closed section always lies within the section and outside the section for an open section member.
- ④ Since, load cannot be applied to the shear centre, the load can be equated to a shear load at the shear centre and torque around shear centre and solutions can be superimposed.

Elastic Axis:

Elastic Axis is the locus of all sections, hence, it is defined for a beam, and goes from tip to root of the beam. It is the ~~root~~ locus of all shear centres of each section along the beam.



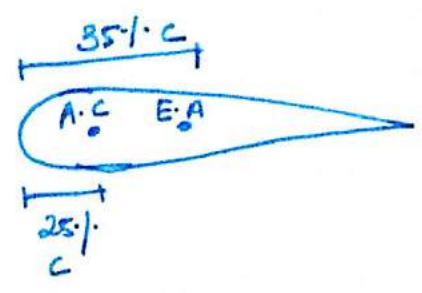
A shear load applied at section A-A at the shear centre, the section A-A will undergo only vertical E.A. deflection and no rotation.

Similarly, a torque applied, around the shear centre will cause only torsion without bending.

Since, the beam is Continuous, the load applied at Section B-B at the Shear Centre will cause bending and torsion of Section A-A.

Importance of elastic axis with reference to Aerodynamic Centre of wing.

- The elastic axis is dependent on geometry, skin, stringer, spar location etc. For a wing the elastic axis may lie somewhere between 35% of chord to 50% of chord. Elastic axis need not be straight. This is due to additional equipment on the wing.
- Since, aerodynamic lift acts at the aerodynamic Centre, which is forward of the elastic axis, every section of wing will undergo, bending and torsion. The torque depends on the distance between the aerodynamic Centre and the elastic axis

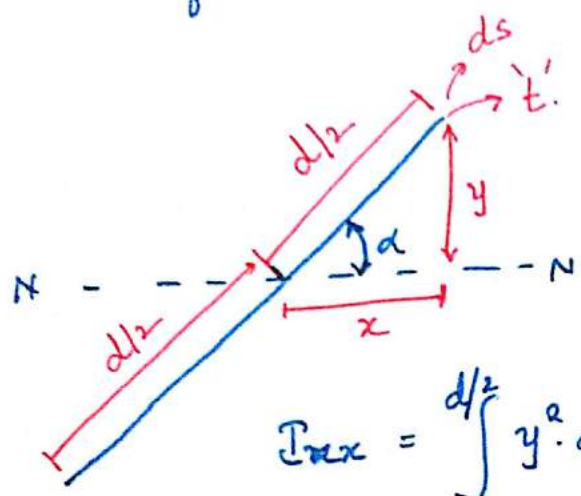
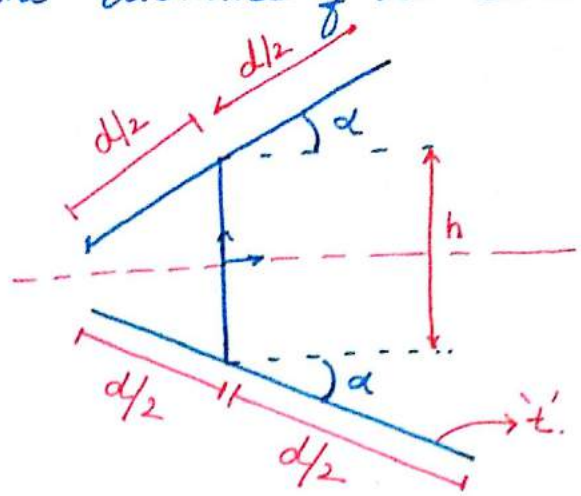


It is because of this that the aircraft wing at every section will undergo bending (vertical deflection) as well as torsion. This is also the cause for few aero-elastic instabilities.

Problem A beam has a Singly Symmetrical thin Walled Cross section as shown. The thickness of the walls is constant throughout, as shown. Find the distance of the shear centre from the web.

Solution:-

Calculation of Moment of Inertia of inclined bodies.



$$\sin \alpha = \frac{y}{d/2}, \quad \frac{d}{2} \cdot \sin \alpha = y \quad \text{--- (1)}$$

$$I_{xx} = \int_0^{d/2} y^2 \cdot dA = \int_0^{d/2} y^2 \cdot t \cdot ds \quad \text{taking } \frac{d}{2} \text{ as } s$$

Substituting from (1),
$$I_{xx} = \int_0^{d/2} \left(\frac{d}{2} \cdot \sin \alpha\right)^2 \cdot t \cdot ds$$

$$I_{xx} = \int_0^{d/2} s^2 \sin^2 \alpha \cdot t \cdot ds, \quad I_{xx} = \sin^2 \alpha \cdot t \int_0^{d/2} s^2 \cdot ds$$

After integrating,
$$I_{xx} = \sin^2 \alpha \cdot t \left[\frac{s^3}{3} \right]_0^{d/2}$$

$$I_{xx} = \sin^2 \alpha \cdot t \cdot \frac{(d/2)^3}{3} = \frac{d^3 \cdot \sin^2 \alpha \cdot t}{24}$$

Hence similarly approach has to be adopted for the other section.

Hence, the total
$$I_{xx} = \frac{2 \cdot t d^3 \sin^2 \alpha}{24}$$

$$I_{xx} = \frac{2 t d^3 \sin^2 \alpha}{12}$$

Similarly, $I_{yy} = \int x^2 dA.$

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$$\cos \frac{\alpha}{2} = \frac{x}{s} = \frac{x}{d/2} \quad \text{or} \quad \frac{d}{2} \cdot \cos \frac{\alpha}{2} = x.$$

$$I_{yy} = \int_0^{d/2} x^2 \cdot t \cdot ds = \int_0^{d/2} \left(\frac{d}{2} \cos \frac{\alpha}{2} \right)^2 \cdot t \cdot ds = \int_0^{d/2} s^2 \cos^2 \alpha \cdot t \cdot ds$$

$$I_{yy} = \cos^2 \alpha \cdot t \cdot \int_0^{d/2} s^2 \cdot ds = \frac{\cos^2 \alpha \cdot t \cdot s^3}{3} \Big|_0^{d/2}$$

$I_{yy} = \frac{\cos^2 \alpha \cdot t \cdot d^3}{24}$, Similarly, the other part of the body can be calculated, hence, we can multiply it by 2.

$$I_{yy} = 2 \cdot \frac{\cos^2 \alpha \cdot t \cdot d^3}{24}, \quad \boxed{I_{yy} = \frac{\cos^2 \alpha \cdot t \cdot d^3}{12}}$$

Similarly, Calculating the cross product of inertia, I_{xy} .

$$I_{xy} = 2 \int_0^{d/2} (x)(y) \cdot dA = 2 \int_0^{d/2} (x)(y) \cdot t \cdot ds.$$

$$I_{xy} = 2 \int_0^{d/2} (s \cos \alpha)(s \sin \alpha) \cdot t \cdot ds = 2 \sin \alpha \cos \alpha \cdot t \int_0^{d/2} s^2 \cdot ds$$

$$I_{xy} = 2 \sin \alpha \cos \alpha \cdot t \cdot \frac{s^3}{3} \Big|_0^{d/2} = \frac{2 \sin \alpha \cos \alpha \cdot d^3 \cdot t}{24} = \frac{2 \sin \alpha \cos \alpha \cdot d^3 \cdot t}{12}.$$

$$\boxed{I_{xy} = \frac{\sin \alpha \cos \alpha \cdot d^3 \cdot t}{12}}$$

Now, solving for the distance of the shear centre,

Since, the cross section is symmetric about x , $I_{xy}=0$
and also since S_x is not present $S_x=0$.

The equation,

$$Q_s = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int t \cdot x \cdot ds - \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int t \cdot y \cdot ds$$

which reduces to,

$$Q_s = - \frac{S_y I_{yy}}{I_{xx} I_{yy}} \int t \cdot y \cdot ds = - \frac{S_y}{I_{xx}} \int t \cdot y \cdot ds$$

Calculating I_{xx} , $A_1 = td = A_3$, $A_2 = ht$

$$I_{xx_1} = \frac{td^3 \sin^2 \alpha}{12} = I_{xx_3} \quad \text{and} \quad I_{xx_2} = \frac{th^3}{12}$$

Hence, $I_{xx} = \left[\frac{td^3 \sin^2 \alpha}{12} + td(h/2)^2 \right] + \left[\frac{th^3}{12} + ht(0)^2 \right] + \left[\frac{td^3 \sin^2 \alpha}{12} + td(h/2)^2 \right]$

$$I_{xx} = 2 \left[\frac{td^3 \sin^2 \alpha}{12} + \frac{tdh^2}{4} \right] + \left[\frac{th^3}{12} \right]$$

$$I_{xx} = \frac{td^3 \sin^2 \alpha}{6} + \frac{tdh^2}{2} + \frac{th^3}{12}, \quad I_{xx} = \left[\frac{td^3 \sin^2 \alpha}{6} + \frac{tdh^2}{2} \right] + \frac{th^3}{12}$$

$$I_{xx} = \frac{th^3}{12} + \frac{th^3}{12} \left[\frac{d^3}{h^3} \cdot \frac{\sin^2 \alpha}{6} + \frac{d}{h} \cdot \frac{6}{2} \right], \quad \text{taking } d/h \text{ as } \gamma \text{ (gamma)}$$

$$I_{xx} = \frac{th^3}{12} \left[1 + \frac{\gamma^3 \sin^2 \alpha}{6} + \frac{6\gamma}{2} \right] = \frac{th^3}{12} + \frac{th^3}{12} \left[\frac{\gamma^3 \sin^2 \alpha}{6} + \frac{\gamma}{2} \right]$$

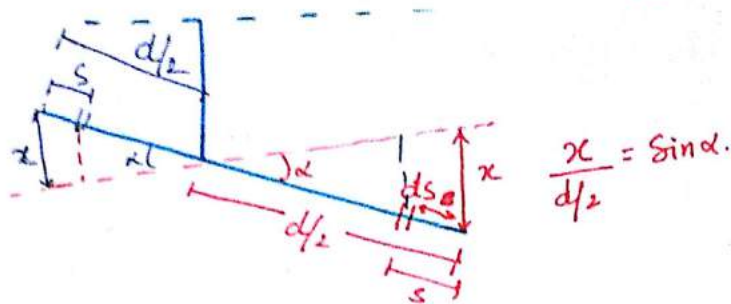
$$I_{xx} = \frac{th^3}{12} + \frac{th^3}{12} \left[\frac{2\gamma^3 \sin^2 \alpha + 6\gamma}{12} \right] = \frac{th^3}{12} + \frac{th^3}{12} \left[\frac{2\gamma^3 \sin^2 \alpha + 6\gamma}{12} \right]$$

$$I_{xx} = \frac{th^3}{12} \left[1 + 2\gamma^3 \sin^2 \alpha + 6\gamma \right] //$$

$$Q_{12} = \frac{-S_y}{I_{xx}} \int_0^s t \cdot y \cdot ds$$

$$Q_{12} = \frac{-S_y \cdot t}{I_{xx}} \int_0^s \left[-\frac{h}{2} - \left(\frac{d}{2} - s \right) \sin \alpha \right] \cdot ds$$

$$Q_{12} = \frac{S_y \cdot t}{I_{xx}} \int_0^s \left[\frac{h}{2} + \left(\frac{d}{2} - s \right) \sin \alpha \right] \cdot ds$$



Hence, at element ds

$$\frac{x}{(d/2)} = \sin \alpha, \quad x = \sin \alpha \left(\frac{d}{2} - s \right)$$

Similarly at 2-3

$$Q_{23} = -\frac{S_y \cdot t}{I_{xx}} \int_0^s t \cdot y \cdot ds$$

$$Q_{23} = -\frac{S_y \cdot t}{I_{xx}} \int_0^s \left[-\frac{h}{2} + \left(\frac{d}{2} - s \right) \sin \alpha \right] \cdot ds, \quad Q_{23} = +\frac{S_y \cdot t}{I_{xx}} \int_0^s \left[+\frac{h}{2} - \left(\frac{d}{2} - s \right) \sin \alpha \right] \cdot ds$$

Integrating Q_{12} and Q_{23} , we get.

$$Q_{12} = \frac{S_y \cdot t}{I_{xx}} \left[\frac{hs}{2} + \left(\frac{d}{2} - s \right) \frac{ds}{2} \sin \alpha - \frac{s^2}{2} \sin \alpha \right]_0^s$$

$$Q_{12} = \frac{S_y \cdot t}{2 I_{xx}} \left[hs + d \cdot s \sin \alpha - s^2 \sin \alpha \right]$$

and by integrating Q_{23} ,

$$Q_{23} = \frac{S_y \cdot t}{2 I_{xx}} \left[hs - \frac{d \cdot s}{\sin \alpha} + s^2 \sin \alpha \right]_0^s$$

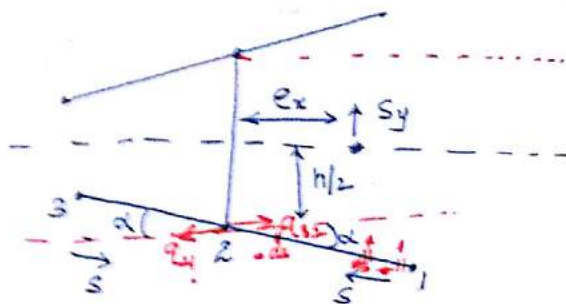
$$Q_{23} = \frac{S_y \cdot t}{2 I_{xx}} \left[hs - d \cdot s \sin \alpha + s^2 \sin \alpha \right]$$

Taking Moments about a point on the web.

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$$S_y e_x = 2 \left[- \int_0^{d/2} q_{12} (ds \cos \alpha) \cdot \frac{h}{2} + \int_0^{d/2} q_{32} (ds \cos \alpha) \cdot \frac{h}{2} \right]$$

Small element over 1-2



or. $S_y \cdot e_x =$

$$2 \left[- \int_0^{d/2} q_{12} \cos \alpha \cdot \frac{h}{2} ds + \int_0^{d/2} q_{32} \cos \alpha \cdot \frac{h}{2} ds \right]$$

Integrating,

$$S_y e_x = 2 \left[- \int_0^{d/2} \left[\frac{S_y t}{2 I_{xx}} \left[h s + d \cdot s \cdot \sin \alpha - s^2 \sin \alpha \right] (ds \cos \alpha) \cdot \frac{h}{2} \right. \right. \\ \left. \left. + \int_0^{d/2} \left[\frac{S_y t}{2 I_{xx}} \left[h s - d \cdot s \sin \alpha + s^2 \sin \alpha \right] (ds \cos \alpha) \cdot \frac{h}{2} \right] \right]$$

$$\Rightarrow S_y \cdot e_x = \frac{S_y \cdot t}{I_{xx}} \left[\left(\frac{h s^2}{2} - \frac{d \cdot s^2}{2} \sin \alpha + \frac{s^3}{3} \sin \alpha \right) \Big|_0^{d/2} + \left(\frac{h s^2}{2} - \frac{d \cdot s^2}{2} \sin \alpha + \frac{s^3}{3} \sin \alpha \right) \Big|_0^{d/2} \right] \times \left[\cos \alpha \cdot \frac{h}{2} \right]$$

$$e_x = \frac{t \cdot \cos \alpha \cdot h}{2 I_{xx}} \left[- \frac{d s^2}{2} \sin \alpha + \frac{2 s^3}{3} \sin \alpha \right] \Big|_0^{d/2}$$

Applying limits,

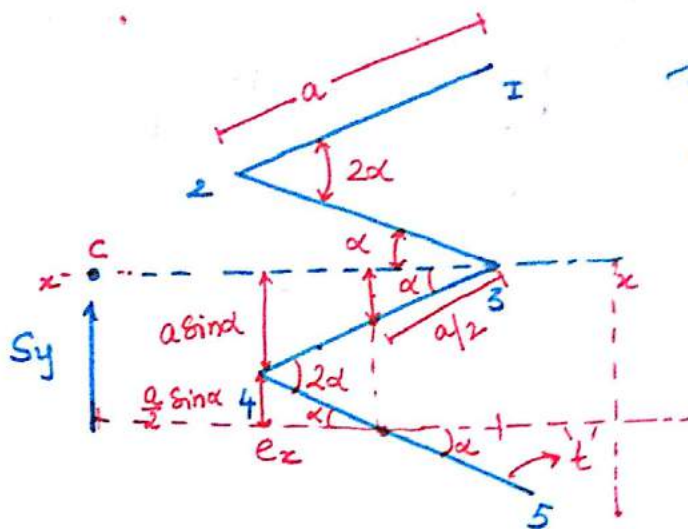
$$e_x = \frac{t \cdot \cos \alpha \cdot h}{2 I_{xx}} \left[- \frac{d^3}{4} \sin \alpha + \frac{2 \frac{d^3}{8} \cdot \sin \alpha}{3} \right]$$

$$e_x = \frac{t \cdot \cos \alpha \cdot h}{2 I_{xx}} \left[- \frac{d^3}{4} \sin \alpha + \frac{d^3}{12} \sin \alpha \right] = \frac{t \cdot \cos \alpha \cdot h \cdot d^3 \sin \alpha}{2 I_{xx}} \left[\frac{1}{12} - \frac{1}{4} \right]$$

$$e_x = \frac{t h d^3 \sin \alpha \cos \alpha}{2 I_{xx}} \left[- \frac{2}{48} \right] = - \frac{t h d^3 \sin \alpha \cos \alpha}{12 I_{xx}}$$

For further simplification
Substitute I_{xx} .

Find the Shear Centre for the Cross-Section Shown in fig.:



From the figure we can say.

$I_{xy} = 0$ and $S_x = 0$
hence the shear flow equation reduces to,

$$q_s = -\frac{S_y}{I_{xx}} \int_0^s t \cdot y \cdot ds$$

Calculation I_{xx} , for lower body, 3-4 and 4-5.

$$I_{xx} = \left[\frac{ta^3 \sin^2 \alpha}{12} + at \left[\frac{a \sin \alpha}{2} \right]^2 \right]_{3-4} + \left[\frac{ta^3 \sin^2 \alpha}{12} + at \left[\frac{a \sin \alpha}{2} + \frac{a}{2} \sin \alpha \right]^2 \right]_{4-5}$$

Total I_{xx} will be,

$$I_{xx} = 2 \left[\frac{ta^3 \sin^2 \alpha}{12} + \frac{a^3 t \sin^2 \alpha}{4} + \frac{ta^3 \sin^2 \alpha}{12} + at \left[a^2 \sin^2 \alpha + \frac{a^2}{4} \sin^2 \alpha + a^2 \sin^2 \alpha \right] \right]$$

$$I_{xx} = 2 \left[2 \left(\frac{ta^3 \sin^2 \alpha}{12} \right) + \frac{a^3 t \sin^2 \alpha}{4} + \frac{a^3 t \sin^2 \alpha}{4} + \frac{a^3 t \sin^2 \alpha}{4} + a^3 t \sin^2 \alpha \right]$$

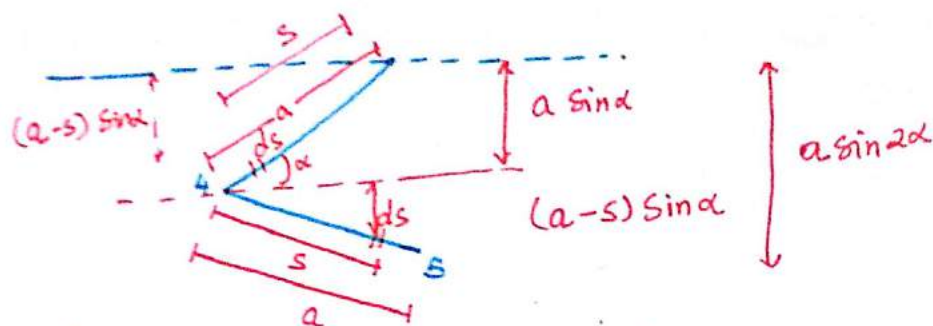
$$I_{xx} = 2 \left[2 \left(\frac{ta^3 \sin^2 \alpha}{12} \right) + \frac{a^3 t \sin^2 \alpha}{2} + 2a^3 t \sin^2 \alpha \right]$$

$$I_{xx} = ta^3 \sin^2 \alpha \left[\frac{1}{3} + 1 + 4 \right] = ta^3 \sin^2 \alpha \left[\frac{4}{3} + 4 \right] = ta^3 \sin^2 \alpha \left[\frac{16}{3} \right]$$

$$I_{xx} = \frac{ta^3 \sin^2 \alpha \cdot 16}{3}$$

Flange 4-5,

(22)



$$Q_{45} = \frac{-S_y}{I_{xx}} \int_0^s t \cdot y \cdot ds$$

$$y = -[a \sin \alpha + (a-s) \sin \alpha]$$

$$y = -[a \sin \alpha + a \sin \alpha - s \sin \alpha]$$

$$y = -[2a \sin \alpha - s \sin \alpha]$$

$$y = (s - 2a) \sin \alpha$$

$$Q_{45} = \frac{-S_y}{I_{xx}} \int_0^s t \cdot (s - 2a) \sin \alpha \cdot ds$$

$$Q_{45} = \frac{S_y \cdot t \cdot \sin \alpha}{I_{xx}} \left[2as - \frac{2a s^2}{2} \right]_0^s = \frac{S_y \cdot t \cdot \sin \alpha}{I_{xx}} \left[2as - \frac{s^2}{2} \right]$$

$$Q_{45} \quad y = (a-s) \sin \alpha, \quad y = (s-a) \sin \alpha$$

Q_{45} need not be calculated as it passes through the point 'S' about which the moments will be taken.

$$S_y \cdot e_x = \left[\int_0^a -Q_{45} \cdot a \sin 2\alpha \cdot ds \right] \times 2$$

$$S_y \cdot e_x = -2 \int_0^a \frac{S_y \cdot t \cdot \sin \alpha}{I_{xx}} \cdot a \sin 2\alpha \cdot ds \left[2as - \frac{s^2}{2} \right]$$

$$S_y \cdot e_x = -\frac{S_y \cdot t \cdot \sin \alpha \cdot a \sin 2\alpha}{t a^3 \sin^2 \alpha \cdot 16} \cdot 3 \times 2 \int_0^a \left[2as - \frac{s^2}{2} \right] ds$$

$$S_y \cdot e_x = \frac{2 S_y \cdot t \cdot \sin \alpha \cdot a \sin 2\alpha \cdot 3}{t \cdot a^3 \sin^2 \alpha \cdot 16} \left[\frac{2as^2}{2} - \frac{s^3}{6} \right]_0^a$$

$$S_y \cdot e_x = \frac{2 S_y \cdot \sin \alpha \cdot a \sin 2\alpha \cdot 3}{t \cdot a^3 \sin^2 \alpha \cdot 16} \left[\frac{2a^3}{2} - \frac{a^3}{6} \right] = \left[\frac{6a^3 - a^3}{6} \right]$$

$$e_x = \frac{2 \sin \alpha \cdot a \sin 2\alpha \cdot 3}{a^3 \sin^2 \alpha \cdot 16} \left[\frac{6a^3 - a^3}{6} \right] = \frac{\sin \alpha \cdot a \sin 2\alpha}{\sin^2 \alpha \cdot 16} \left[\frac{5}{6} \right]$$

$$e_x = \frac{\sin \alpha \cdot a \sin 2\alpha \cdot 5}{\sin^2 \alpha \cdot 16} = \frac{a \sin 2\alpha \cdot 5}{16 \cdot \sin \alpha}$$

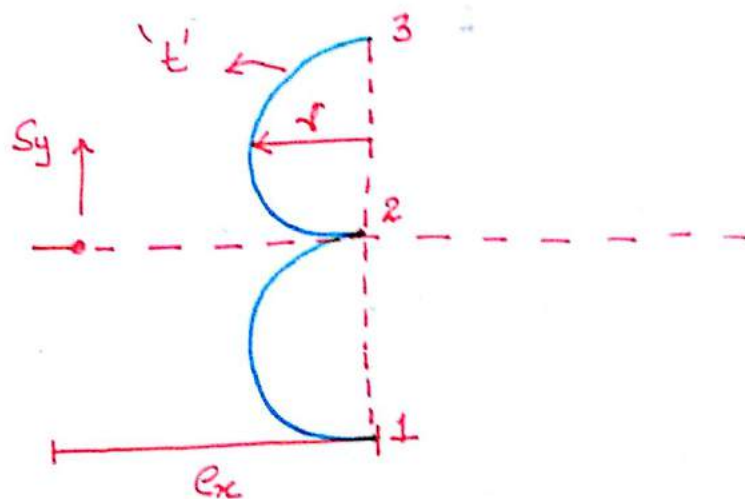
(23)

$$e_x = \frac{2 \cdot a \cdot \sin \alpha \cos \alpha \cdot 5}{\sin^2 \alpha \cdot 16}$$

$$e_x = \frac{2 \cdot a \cdot \cos \alpha \cdot 5}{16}$$

$$e_x = \frac{a \cdot \cos \alpha \cdot 5}{8} //$$

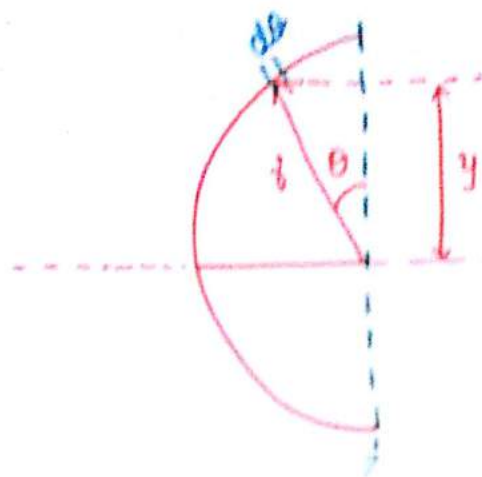
Problem: For the Cross-Section shown in figure. Calculate the offset of Shear Centre e_x .



As it is a Symmetric Cross-section, $I_{xy} = 0$ and $S_x = 0$ hence, the Shear flow equation reduces to,

$$q_s = \frac{-S_y}{I_{xx}} \int_0^s t \cdot y \cdot ds.$$

Calculation of Moments of Inertia for a Circular Cross has to be made in order to find the location of the Shear Centre.



$$I_{xx} = \int_0^\pi y^2 \cdot t \cdot ds$$

$$I_{xx} = \int_0^\pi r^2 \cos^2 \theta \cdot t \cdot r \cdot d\theta$$

$$I_{xx} = r^3 \cdot t \int_0^\pi \cos^2 \theta \cdot d\theta$$

$$y = \cos \theta \cdot r$$

$$I_{xx} = r^3 t \int_0^\pi \frac{1 + \cos 2\theta}{2} \cdot d\theta$$

$$I_{xx} = r^3 t \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^\pi$$

Applying the limits,

$$I_{xx} = r^3 t \cdot \left[\frac{\pi}{2} \right] = \frac{r^3 t \cdot \pi}{2} = I_{xx}$$

Similarly, we can solve for I_{yy} , $I_{yy} = \frac{r^3 t \cdot \pi}{2}$

Solving for the Cross product of Inertia,

$$I_{xy} = \int_0^\pi x \cdot y \cdot t \cdot r \cdot d\theta \quad \text{where } x = r$$

$$I_{xy} = \int_0^\pi r \cdot r \cdot \cos \theta \cdot t \cdot r \cdot d\theta$$

$$I_{xy} = r^3 \cdot t \int_0^\pi \cos \theta \cdot d\theta$$

$$I_{xy} = r^3 \cdot t \left[\sin \theta \cdot d\theta \right]_0^\pi$$

$I_{xy} = 0$ // Because a Circular Section is always Symmetric.

Solving for total I_{xx} .

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$$I_{xx} = 2 \left[I_{xx} + Ay^2 \right]$$

$$I_{xx} = 2 \left[\frac{tr^3\pi}{2} + \pi r t (r^2) \right]$$

$$I_{xx} = 2 \left[\frac{tr^3\pi}{2} + \pi r^3 t \right]$$

$$I_{xx} = 2 \left[\frac{3tr^3\pi}{2} \right]$$

$I_{xx} = 3tr^3\pi$. Hence, the equation becomes,

$$q_{12} = \frac{-S_y}{I_{xx}} \int_0^\theta t [-r - r \cos \theta] r d\theta.$$

$$q_{12} = \frac{S_y}{I_{xx}} \int_0^\theta tr [\theta + r \cos \theta] d\theta$$

$$q_{12} = \frac{S_y \cdot tr}{3tr^3\pi} \int_0^\theta \theta + r \cos \theta \cdot d\theta, \text{ Integrating, we get,}$$

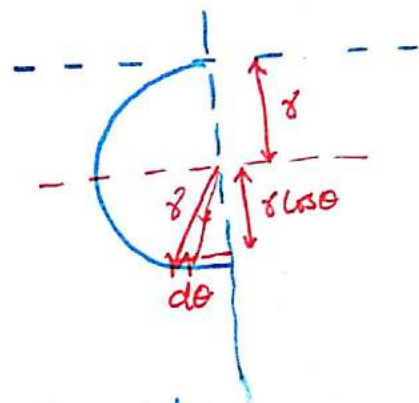
$$q_{12} = \frac{S_y}{3r^2\pi} [\theta + r \sin \theta]_0^\theta, \quad q_{12} = \frac{S_y}{3r^2\pi} [\theta + r \sin \theta]$$

Taking moments about point 2.

$$S_y \cdot e_x = 2 \int_0^\pi q_{12} [r + r \cos \theta] r \cdot d\theta$$

$$S_y \cdot e_x = \frac{2 \cdot S_y \cdot r^2}{3r^2\pi} \int_0^\pi [1 + \cos \theta] [\theta + r \sin \theta] d\theta$$

$$e_x = \frac{2}{3} \frac{S_y \cdot r}{\pi} \int_0^\pi [1 + \cos \theta] [\theta + r \sin \theta] d\theta.$$



Assuming both are acting in the Anti-clockwise direction, the negative signs cancel out.

$$e_x = \frac{2\gamma}{3\pi} \int_0^{\pi} [\theta + \sin\theta + \theta \cos\theta + \sin\theta \cos\theta] d\theta$$

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~~$e_x = \frac{2\gamma}{3\pi} \left[\frac{\theta^2}{2} + \cos\theta + \theta \sin\theta + \frac{\sin^2\theta}{2} \right]_0^{\pi}$~~ , $\sin\theta$ and $\cos\theta$ would cancel out when we integrate and apply limits.

$$e_x = \frac{2\gamma}{3\pi} \left[\frac{\theta^2}{2} \right]_0^{\pi} = \frac{2\gamma}{3\pi} \left[\frac{\pi^2}{2} - 0 \right] = \frac{2\gamma}{3\pi} \left[\frac{\pi^2}{2} \right]$$

$$e_x = \frac{2\pi\gamma}{6} = \boxed{\frac{\pi\gamma}{3} = e_x}$$